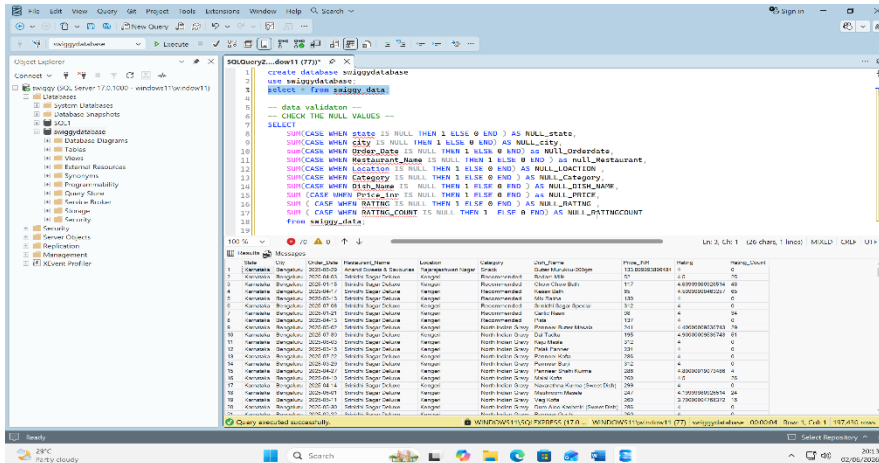


SQL Data Analysis Project

Create database swiggydatabase;
use swiggydatabase;
select * from swiggy_data;

made by pandurang yadav

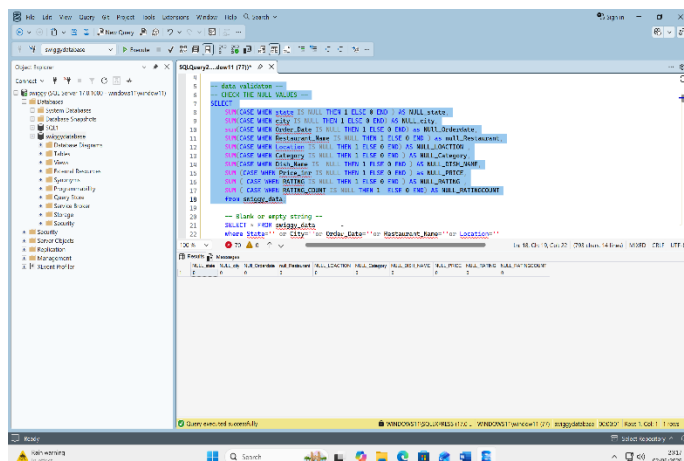


-- DATA VALIDATION --

-- CHECK THE NULL VALUES --

SELECT

SUM(CASE WHEN state IS NULL THEN 1 ELSE 0 END) AS NULL_state,
SUM(CASE WHEN city IS NULL THEN 1 ELSE 0 END) AS NULL_city,
sum(CASE WHEN Order_Date IS NULL THEN 1 ELSE 0 END) as NULL_Orderdate,
SUM(CASE WHEN Restaurant_Name IS NULL THEN 1 ELSE 0 END) as null_Restaurant,
SUM(CASE WHEN Location IS NULL THEN 1 ELSE 0 END) AS NULL_Location ,
SUM(CASE WHEN Category IS NULL THEN 1 ELSE 0 END) AS NULL_Category,
SUM(CASE WHEN Dish_Name IS NULL THEN 1 ELSE 0 END) AS NULL_Dish_Name,
SUM (CASE WHEN Price_inr IS NULL THEN 1 ELSE 0 END) as NULL_PRICE,
SUM (CASE WHEN Rating IS NULL THEN 1 ELSE 0 END) AS NULL_RATING ,
SUM (CASE WHEN Rating_Count IS NULL THEN 1 ELSE 0 END) AS NULL_RatingCount from
swiggy_data;

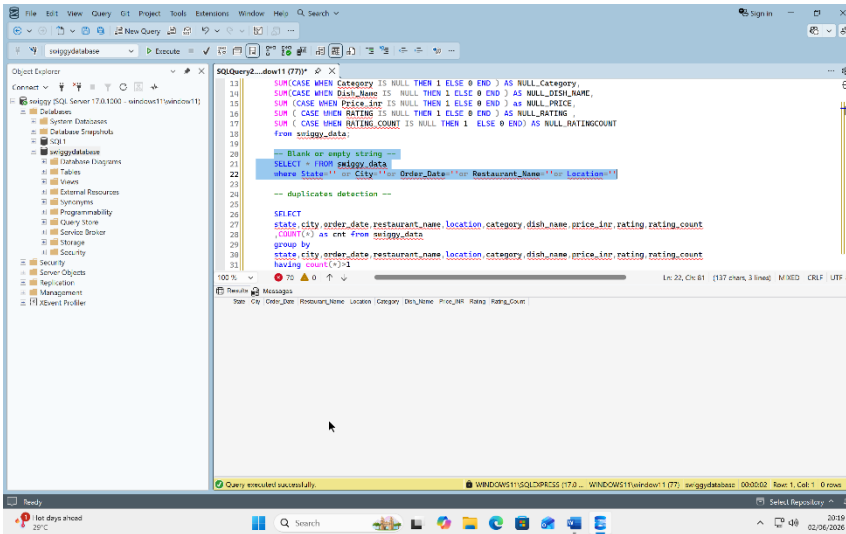


SQL Data Analysis Project

-- BLANK OR EMPTY STRING --

```
SELECT * FROM swiggy_data
```

where State="" or City=""or Order_Date=""or Restaurant_Name=""or Location=""



-- DUPLICATES DETECTION --

SELECT

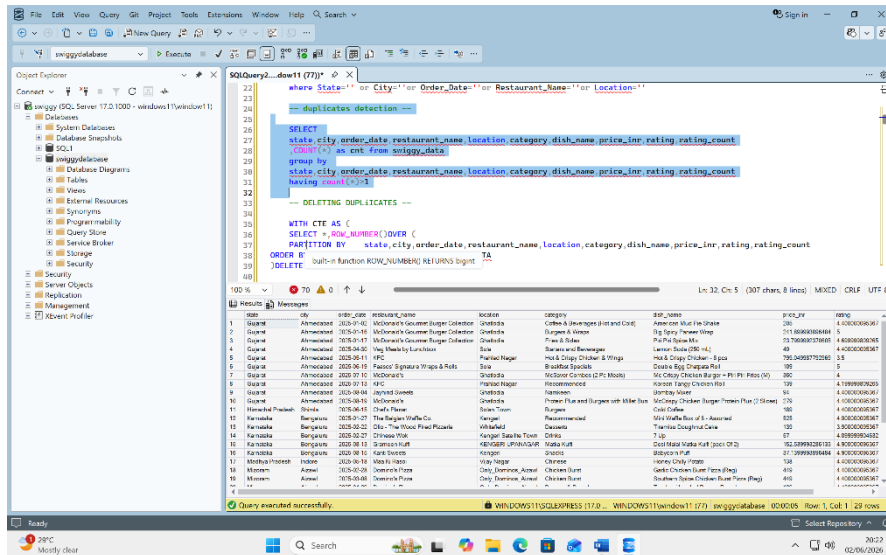
state,city,order_date,restaurant_name,location,category,dish_name,price_inr,rating,rating_count

```
,COUNT(*) as cnt from swiggy_data
```

group by

state,city,order_date,restaurant_name,location,category,dish_name,price_inr,rating,rating_count

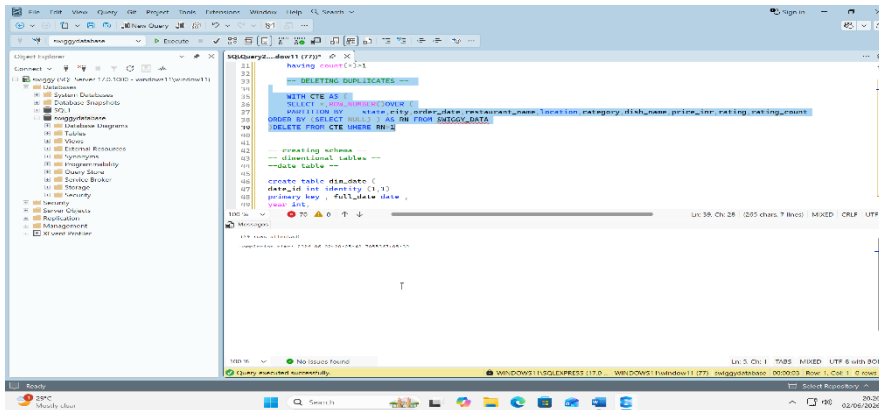
having count(*)>1



```
-- DELETING DUPLICATES --
```

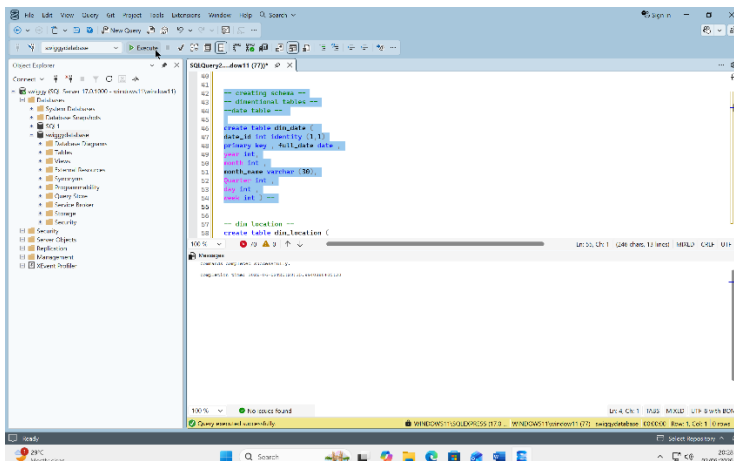
SQL Data Analysis Project

```
WITH CTE AS (  
SELECT *,ROW_NUMBER()OVER (  
PARTITION BY  
state,city,order_date,restaurant_name,location,category,dish_name,price_inr,rating,rating_count  
ORDER BY (SELECT NULL) ) AS RN FROM SWIGGY_DATA)  
DELETE FROM CTE WHERE RN>1
```



```
-- CREATING SCHEMA --  
-- DIMENTIONAL TABLES --  
--DATE TABLE --
```

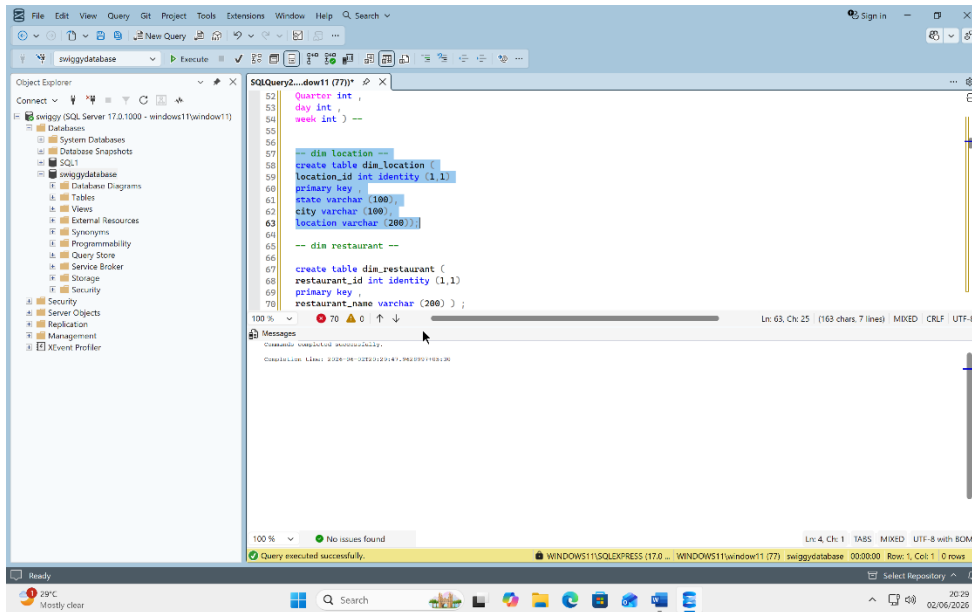
```
create table dim_date (  
date_id int identity (1,1)  
primary key , full_date date ,  
year int ,  
month int ,  
month_name varchar (30),  
Quarter int ,  
day int ,week int )
```



SQL Data Analysis Project

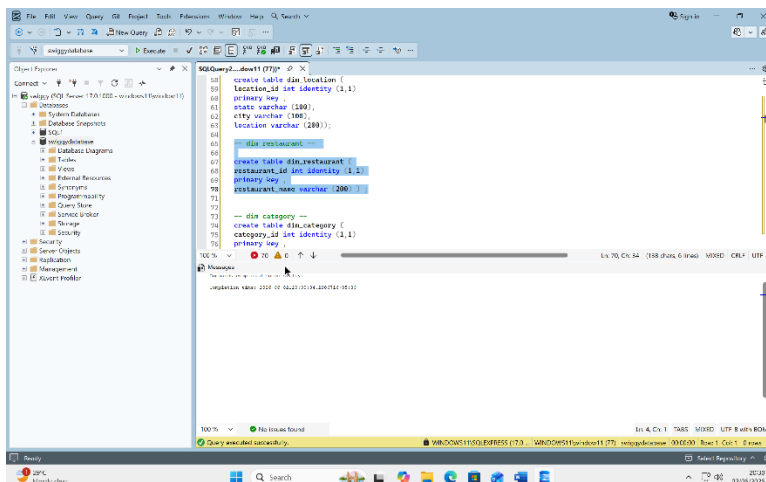
-- DIM LOCATION --

create table dim_location (
location_id int identity (1,1)
primary key ,
state varchar (100),
city varchar (100),
location varchar (200));



-- DIM RESTAURANT --

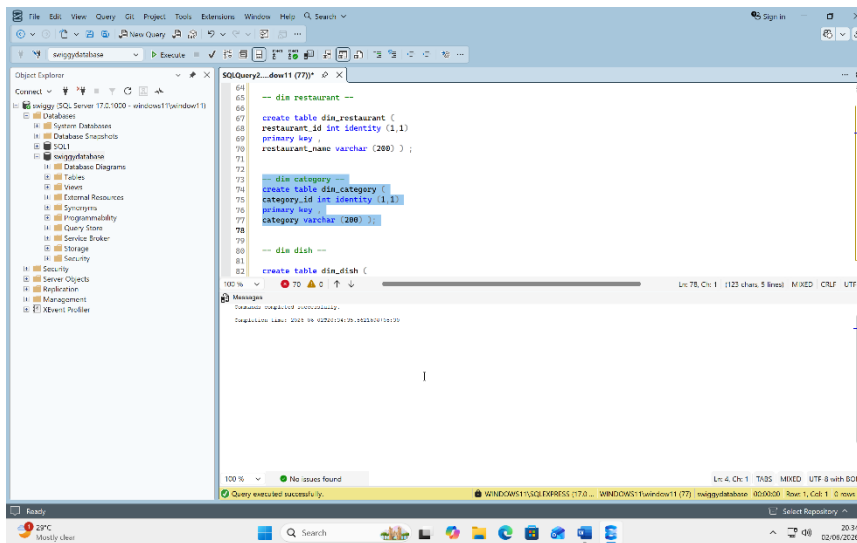
create table dim_restaurant (
restaurant_id int identity (1,1)
primary key ,
restaurant_name varchar (200));



SQL Data Analysis Project

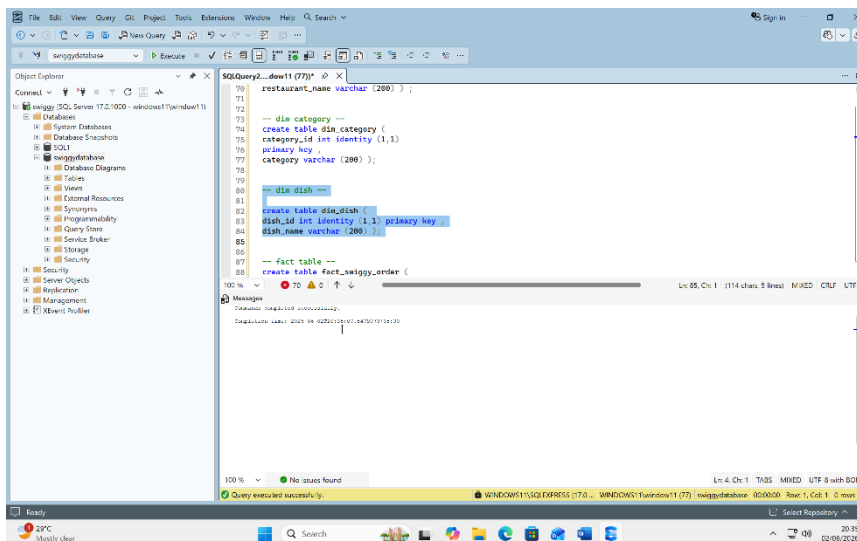
-- dim category --

create table dim_category (
category_id int identity (1,1)
primary key ,
category varchar (200));



-- dim dish --

create table dim_dish (
dish_id int identity (1,1) primary key ,
dish_name varchar (200));



SQL Data Analysis Project

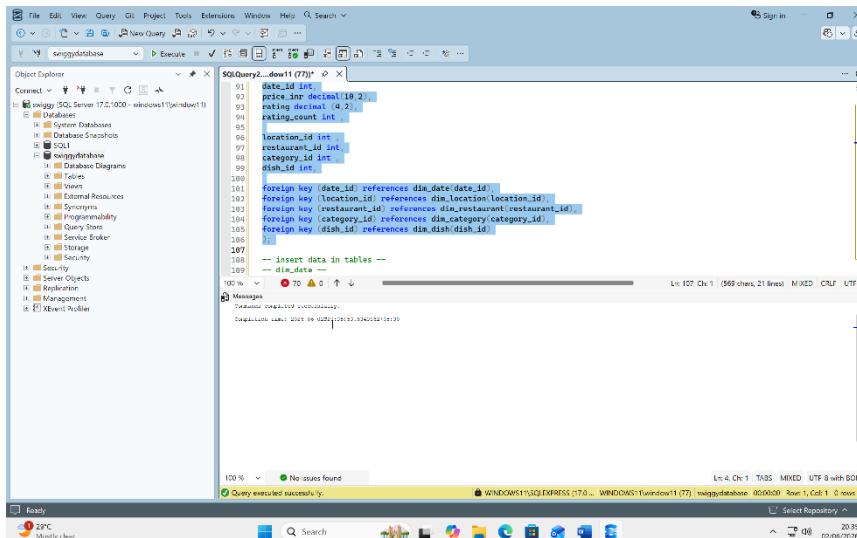
-- fact table --

```
create table fact_swiggy_order (  
order_id int identity (1,1) primary key ,
```

```
date_id int,  
price_inr decimal(10,2),  
rating decimal (4,2),  
rating_count int ,
```

```
location_id int ,  
restaurant_id int,  
category_id int ,  
dish_id int,
```

```
foreign key (date_id) references dim_date(date_id),  
foreign key (location_id) references dim_location(location_id),  
foreign key (restaurant_id) references dim_restaurant(restaurant_id),  
foreign key (category_id) references dim_category(category_id),  
foreign key (dish_id) references dim_dish(dish_id)  
);
```



-- insert data in tables --

-- dim_date --

```
insert into dim_date( full_date,year,month,month_name,Quarter,day,week)
```

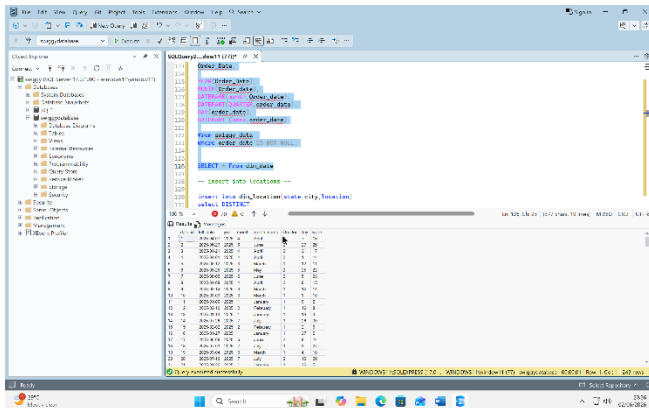
SQL Data Analysis Project

```
select DISTINCT  
Order_Date,
```

```
YEAR(Order_Date),  
MONTH(Order_date),  
DATENAME(month,Order_date),  
DATEPART(QUARTER,order_date),  
DAY(order_date),  
DATEPART (week,order_date)
```

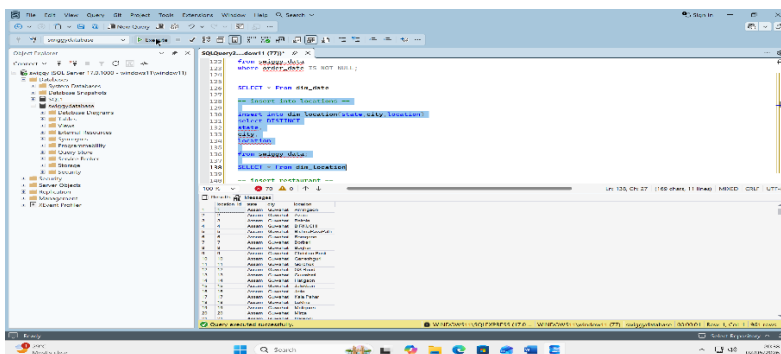
```
from swiggy_data  
where order_date IS NOT NULL;
```

```
SELECT * From dim_date
```



```
-- insert into locations --
```

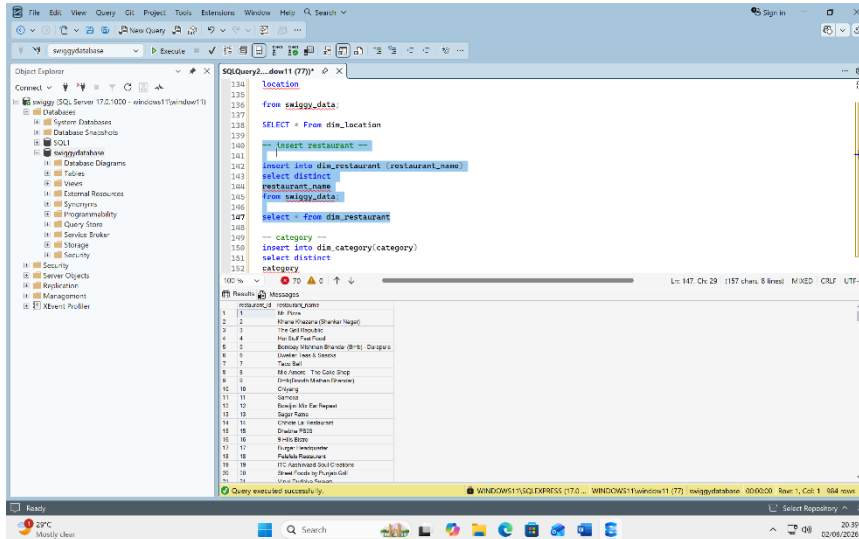
```
insert into dim_location(state,city,location)  
select DISTINCT  
state,  
city,  
location  
from swiggy_data;  
SELECT * From dim_location
```



SQL Data Analysis Project

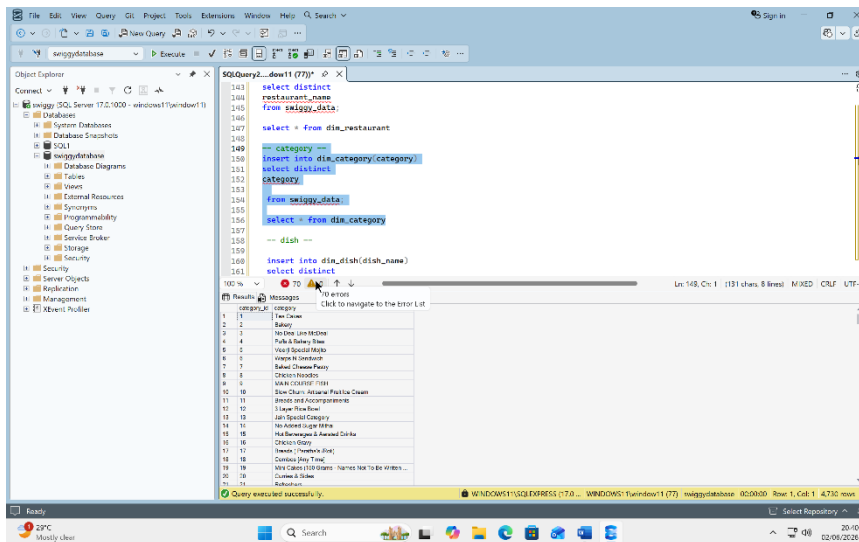
-- insert restaurant --

```
insert into dim_restaurant (restaurant_name)
select distinct
restaurant_name
from swiggy_data;
select * from dim_restaurant
```



-- category --

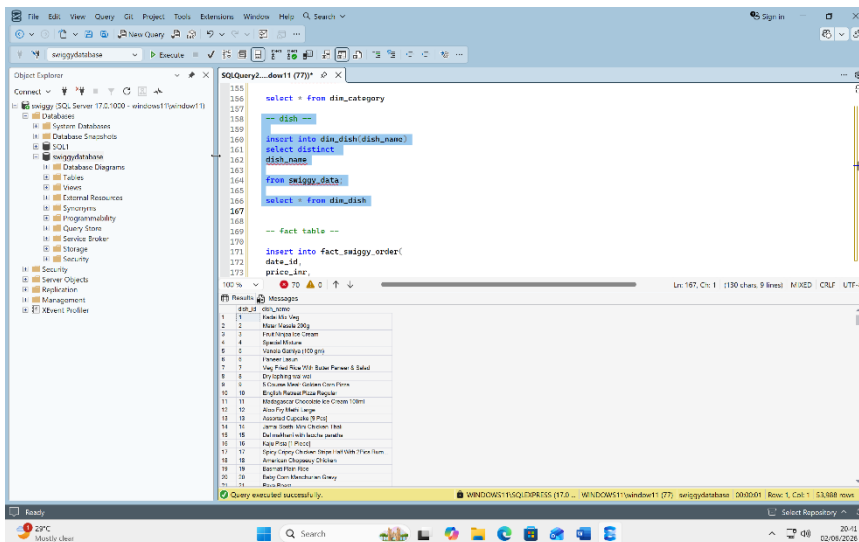
```
insert into dim_category(category)
select distinct
category
from swiggy_data;
select * from dim_category
```



SQL Data Analysis Project

```
-- dish --
```

```
insert into dim_dish(dish_name)
select distinct
dish_name
from swiggy_data;
select * from dim_dish
```



```
-- fact table --
```

```
insert into fact_swiggy_order(
date_id,
price_inr,
rating,
rating_count,
location_id,
restaurant_id,
category_id,
dish_id)
```

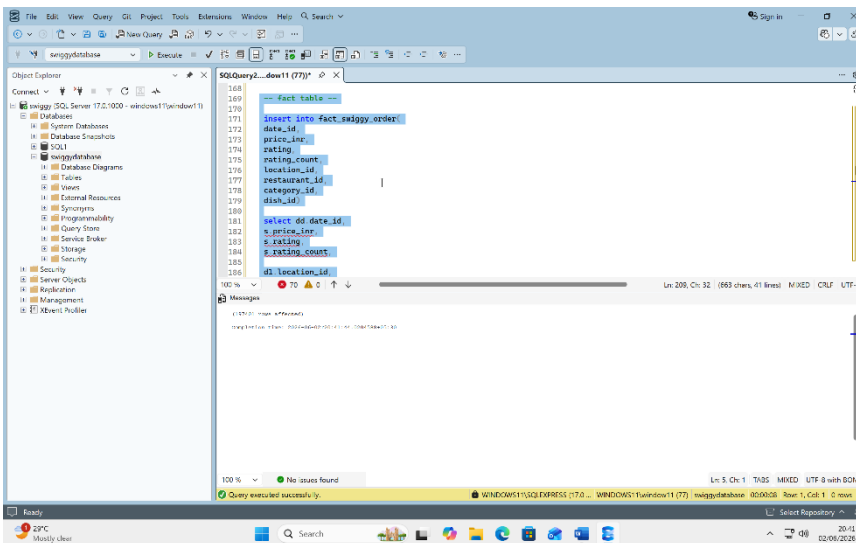
```
select dd.date_id,  
s.price_inr,  
s.rating,  
s.rating_count,
```

```
d1.location_id,  
dr.restaurant_id,  
dc.category_id,  
dsh.dish_id
```

SQL Data Analysis Project

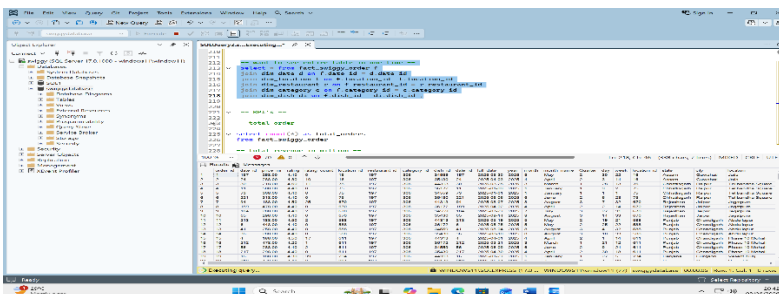
from swiggy_data s

```
join dim_date dd
on dd.full_date = s.Order_Date
join dim_location d1
on d1.state = s.State
and d1.city = s.City
and d1.location = s.Location
join dim_restaurant dr
on dr.restaurant_name = s.Restaurant_Name
join dim_category dc
on dc.category = s.Category
join dim_dish dsh
on dsh.dish_name = s.Dish_Name
```



-- want to see entire table in one time --

```
select * from fact_swiggy_order f
join dim_date d on f.date_id = d.date_id
join dim_location l on f.location_id = l.location_id
join dim_restaurant r on f.restaurant_id = r.restaurant_id
join dim_category c on f.category_id = c.category_id
join dim_dish di on f.dish_id = di.dish_id ;
```

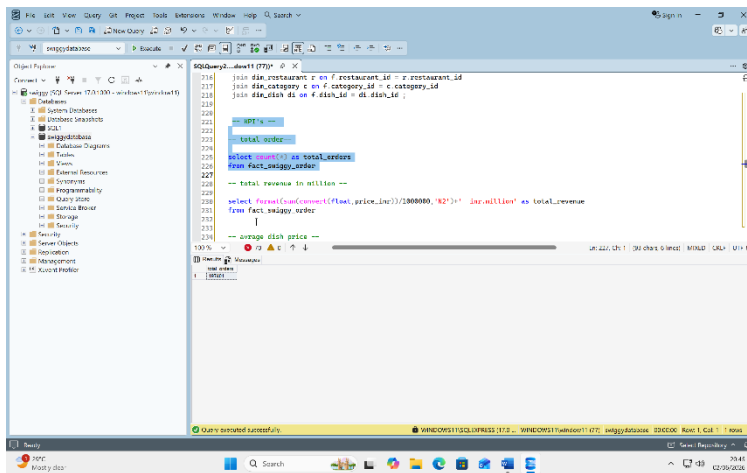


SQL Data Analysis Project

-- KPI's --

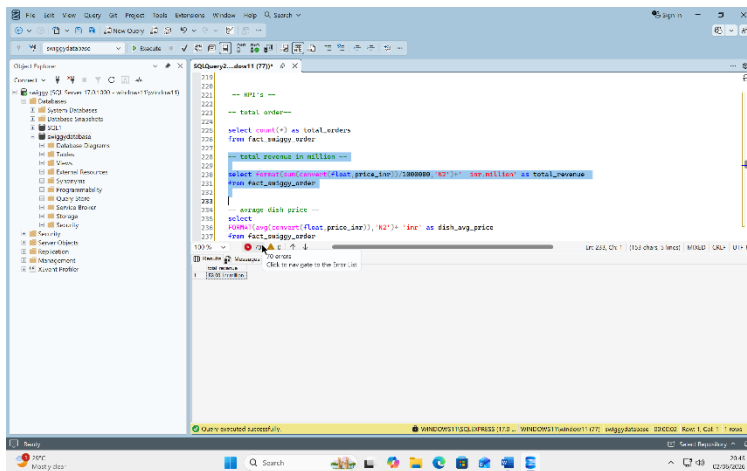
-- total order--

select count(*) as total_orders
from fact_swiggy_order



-- total revenue in million --

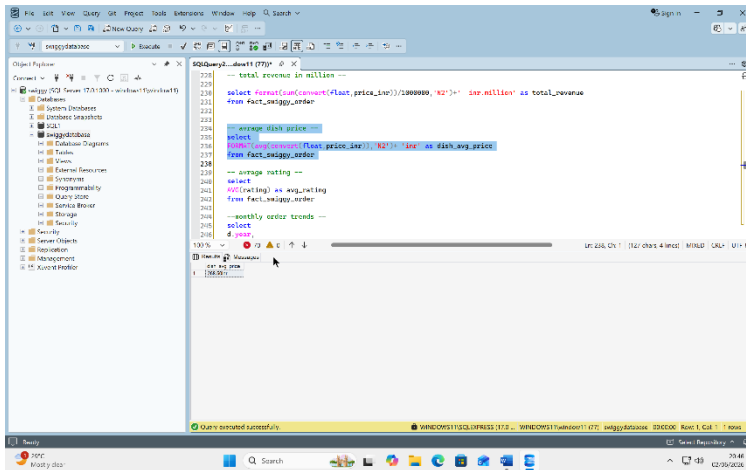
select format(sum(convert(float,price_inr))/1000000,'N2')+ ' inr.million' as total_revenue
from fact_swiggy_order



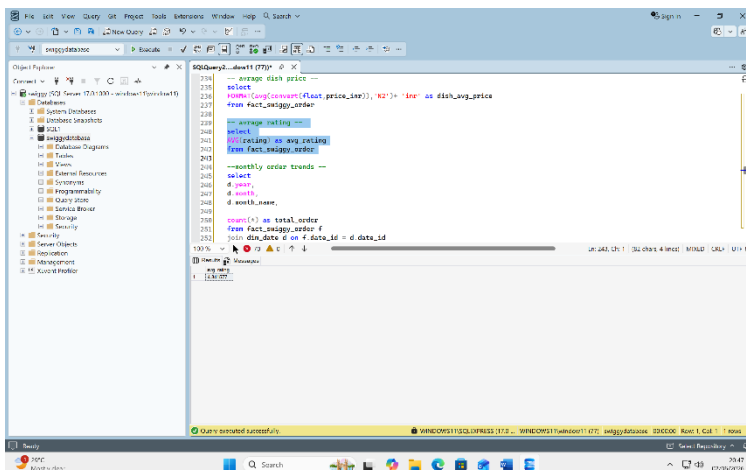
-- avrage dish price --

select
FORMAT(avg(convert(float,price_inr)), 'N2')+ ' inr' as dish_avg_price
from fact_swiggy_order

SQL Data Analysis Project

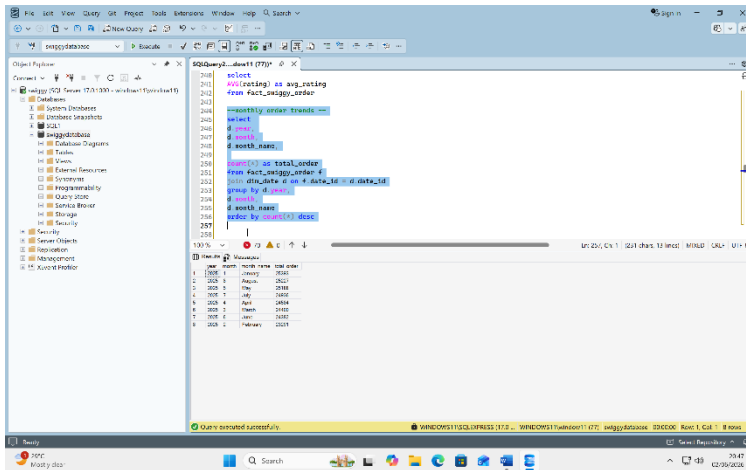


```
-- average rating --
select
AVG(rating) as avg_rating
from fact_swiggy_order
```



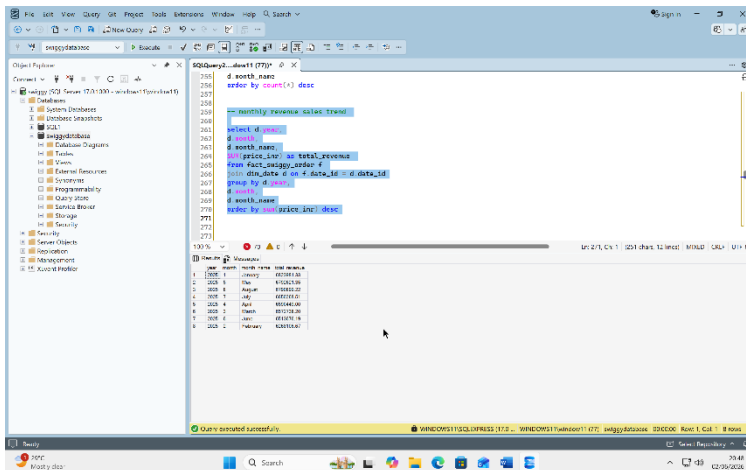
```
--monthly order trends --
select
d.year,
d.month,
d.month_name,
count(*) as total_order
from fact_swiggy_order f
join dim_date d on f.date_id = d.date_id
group by d.year,
d.month,
d.month_name
order by count(*) desc
```

SQL Data Analysis Project



-- monthly revenue sales trend

```
select d.year,
d.month,
d.month_name,
SUM(price_inr) as total_revenue
from fact_swiggy_order f
join dim_date d on f.date_id = d.date_id
group by d.year,
d.month,
d.month_name
order by sum(price_inr) desc
```

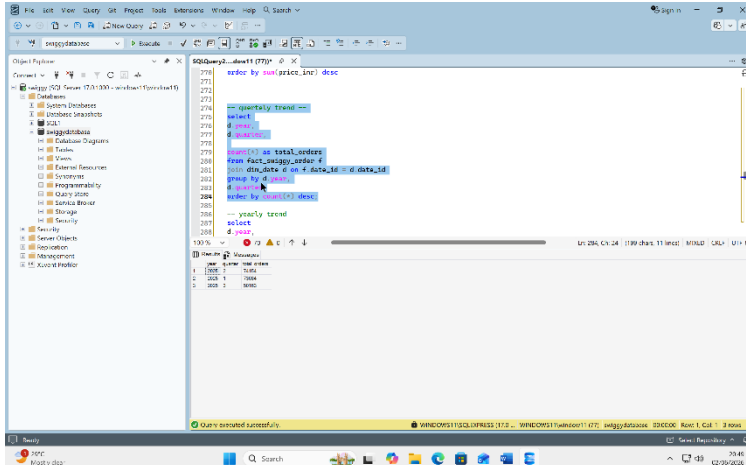


-- quarterly trend --

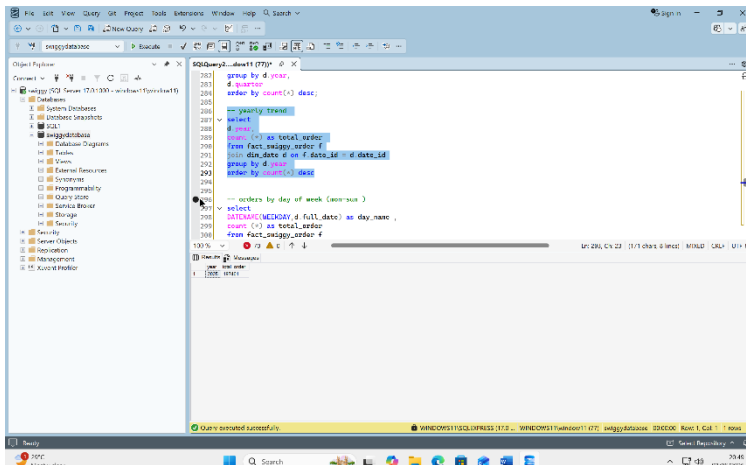
```
select
d.year,
d.quarter,
count(*) as total_orders
```

SQL Data Analysis Project

```
from fact_swiggy_order f
join dim_date d on f.date_id = d.date_id
group by d.year,
d.quarter
order by count(*) desc;
```



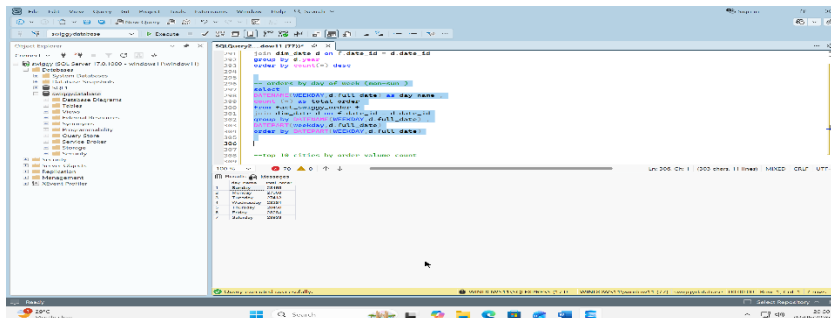
```
-- yearly trend
select
d.year,
count (*) as total_order
from fact_swiggy_order f
join dim_date d on f.date_id = d.date_id
group by d.year
order by count(*) desc
```



```
-- orders by day of week (mon-sun )
select
DATENAME(WEEKDAY,d.full_date) as day_name ,
count (*) as total_order
```

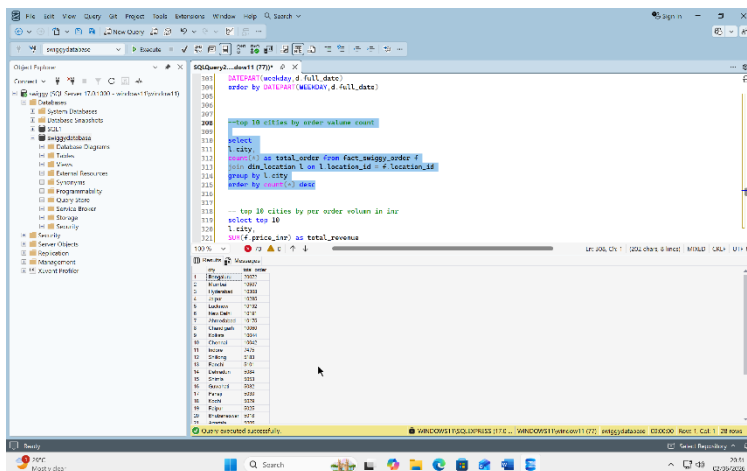
SQL Data Analysis Project

```
from fact_swiggy_order f
join dim_date d on f.date_id = d.date_id
group by DATENAME(WEEKDAY,d.full_date) ,
DATEPART(weekday,d.full_date)
order by DATEPART(WEEKDAY,d.full_date)
```



--top 10 cities by order volume count

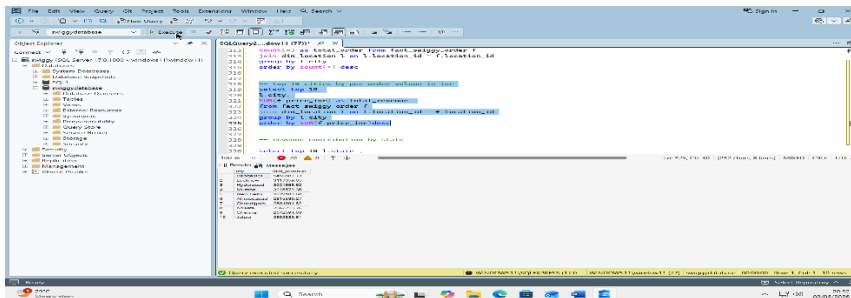
```
select
l.city,
count(*) as total_order from fact_swiggy_order f
join dim_location l on l.location_id = f.location_id
group by l.city
order by count(*) desc
```



-- top 10 cities by per order volumn in inr

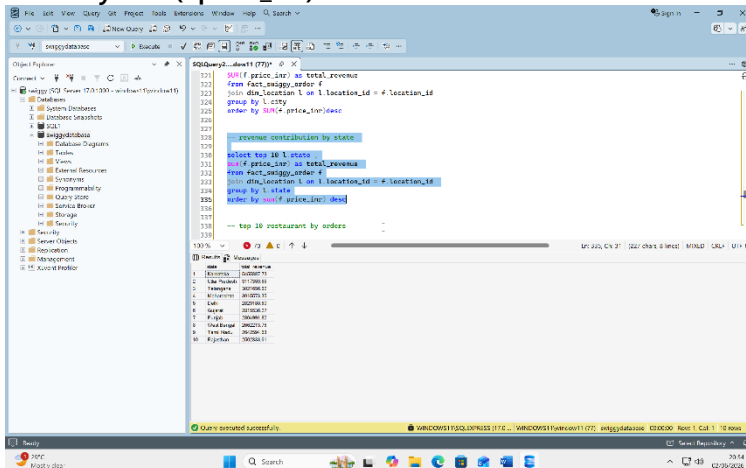
```
select top 10
l.city,
SUM(f.price_inr) as total_revenue
from fact_swiggy_order f
join dim_location l on l.location_id = f.location_id
group by l.city
order by SUM(f.price_inr)desc
```

SQL Data Analysis Project



-- revenue contribution by state

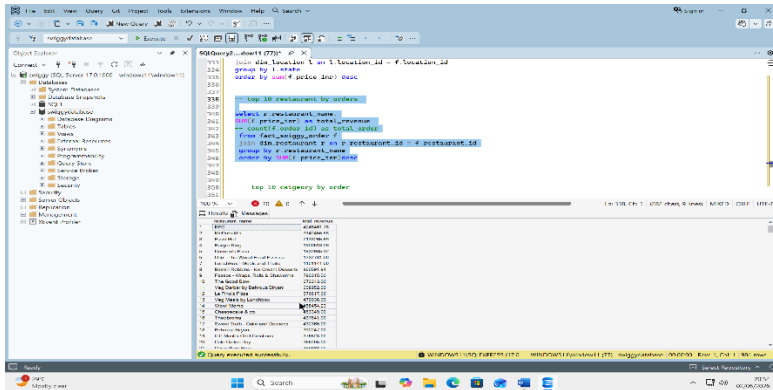
```
select top 10 l.state ,
sum(f.price_inr) as total_revenue
from fact_swiggy_order f
join dim_location l on l.location_id = f.location_id
group by l.state
order by sum(f.price_inr) desc
```



-- top 10 restaurant by orders

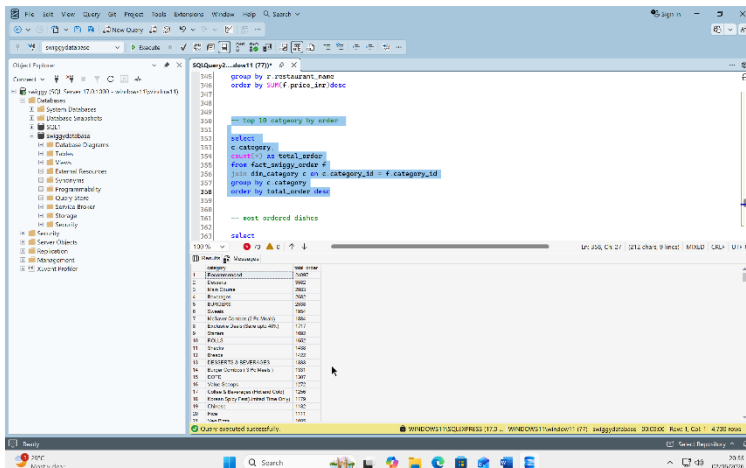
```
select r.restaurant_name,
SUM(f.price_inr) as total_revenue
-- count(f.order_id) as total_order
from fact_swiggy_order f
join dim_restaurant r on r.restaurant_id = f.restaurant_id
group by r.restaurant_name
order by SUM(f.price_inr) desc
```


SQL Data Analysis Project



-- top 10 category by order

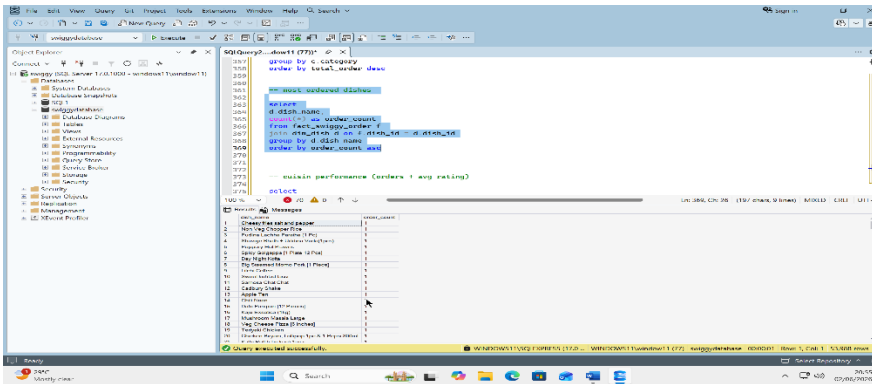
```
select
c.category,
count(*) as total_order
from fact_swiggy_order f
join dim_category c on c.category_id = f.category_id
group by c.category
order by total_order desc
```



-- most ordered dishes

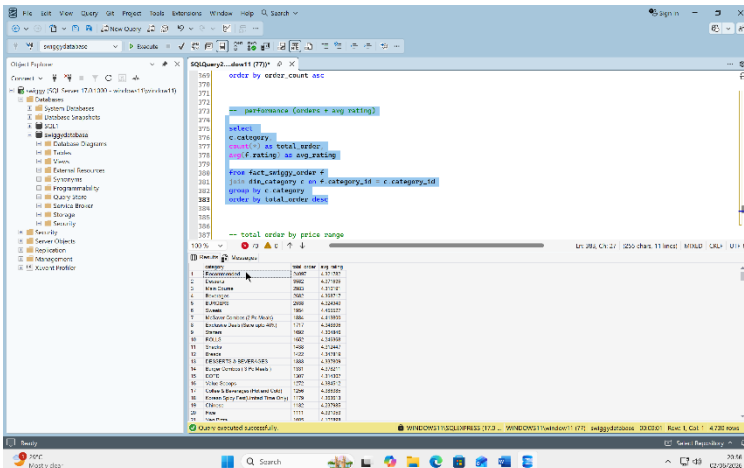
```
select
d.dish_name,
count(*) as order_count
from fact_swiggy_order f
join dim_dish d on f.dish_id = d.dish_id
group by d.dish_name
order by order_count asc
```

SQL Data Analysis Project



-- performance (orders + avg rating)

```
select
c.category,
count(*) as total_order,
avg(f.rating) as avg_rating
from fact_swiggy_order f
join dim_category c on f.category_id = c.category_id
group by c.category
order by total_order desc
```



-- total order by price range

```
select
case
when CONVERT(float,price_inr) < 100 then 'under 100'
when CONVERT(float,price_inr) between 100 and 199 THEN '100-199'
when CONVERT(float,price_inr) between 200 and 299 then '200-299'
when CONVERT(float,price_inr) between 300 and 399 then '300-399'
else '500+'
end as price_range,
count(*) as total_order
from fact_swiggy_order f
```

SQL Data Analysis Project

group by

case

when CONVERT(float,price_inr) < 100 then 'under 100'

when CONVERT(float,price_inr) between 100 and 199 then '100-199'

when CONVERT(float,price_inr) between 200 and 299 then '200-299'

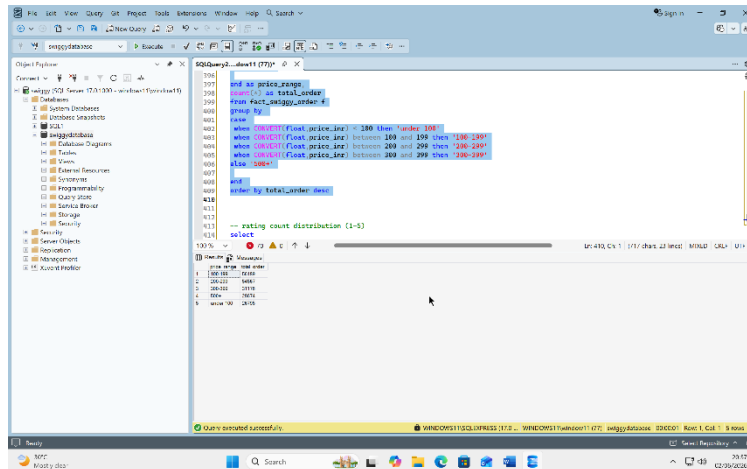
when CONVERT(float,price_inr) between 300 and 399 then '300-399'

```
else '500+'

```

end

```
order by total_order desc
```



-- rating count distribution (1-5)

select

rating,

```
count(*) as rating_count
```

```
from fact_swiggy_order
```

group by rating

order by rating desc

